

Assignment 4

Case Study 1: E-commerce Customer Behavior Trends

An online retail company seeks to understand customer behavior trends to optimize their marketing and website experience. By analyzing customer interactions, the company aims to improve conversions and reduce cart abandonment rates.

Objective:

To identify key patterns in customer behavior that influence purchase decisions and recommend strategies for enhancing the online shopping experience.

Data Overview:

A dataset was collected based on customer activity on the e-commerce platform. The key variables include:

- **Customer Id:**
- **Session Duration:** Time spent on the website (in seconds).
- **Pages Visited:** Number of pages viewed per session.
- **Cart Additions:** Number of products added to the shopping cart.
- **Purchase Made:** Whether a customer completed a purchase.
- **Device Type:** Mobile, Desktop, or Tablet.
- **Referral Source:** The platform through which customers arrived (Google, Facebook, Direct, Email).

Generate the synthetic dataset of 200 samples. Use numpy methods such `random.randint`, `random.choice`. Now apply filtering approaches to analyze customer behavior trends in e-commerce DataFrame.

1. Display customers who made a purchase.
2. Display customers who do not made purchase.
3. Generate list of customers who added items in the cart but did not purchase.
4. Write a code to find percentage of device type used for visiting the pages.
5. Using the same script design in 4, check which referral source used most by the customer.

Case Study 2: Student Mobile Usage Patterns

A university conducted a study to understand mobile usage patterns among students. The goal was to analyze how students use their mobile devices for academic and non-academic purposes and derive insights to enhance digital learning strategies.

Objective:

To evaluate student mobile usage trends, identify key behavioral patterns, and suggest improvements for digital education platforms.

Data Overview:

A dataset was collected from students, capturing various parameters:

- **Student ID:** Unique identifier for each student.
- **Daily Screen Time:** Total time spent on mobile devices (in minutes).
- **Academic Usage (%):** Percentage of screen time spent on educational activities.
- **Social Media Usage (%):** Percentage of time spent on social networking apps.
- **Gaming Usage (%):** Percentage of time dedicated to mobile gaming.
- **Device Type:** The type of mobile device used (Android, iOS).
- **Internet Access Type:** Wi-Fi or Mobile Data.
- **Peak Usage Hours:** monitor usage between 1 AM to 11:30 PM (23:30)

Generate the synthetic dataset of 200 samples. Use numpy methods such `random.randint`, `random.choice`.

Key Insights:

1. **Device Preference:** filter students device preference ratio.
2. **Internet Access Trends:** write python script to find percentage of internet access type.
3. **Peak Usage Hours:** write a python script to check peak mobile usage hours among students.
4. **Academic or Non-Academic usage:** write script to find student mobile usage trend in academic or non-academic.